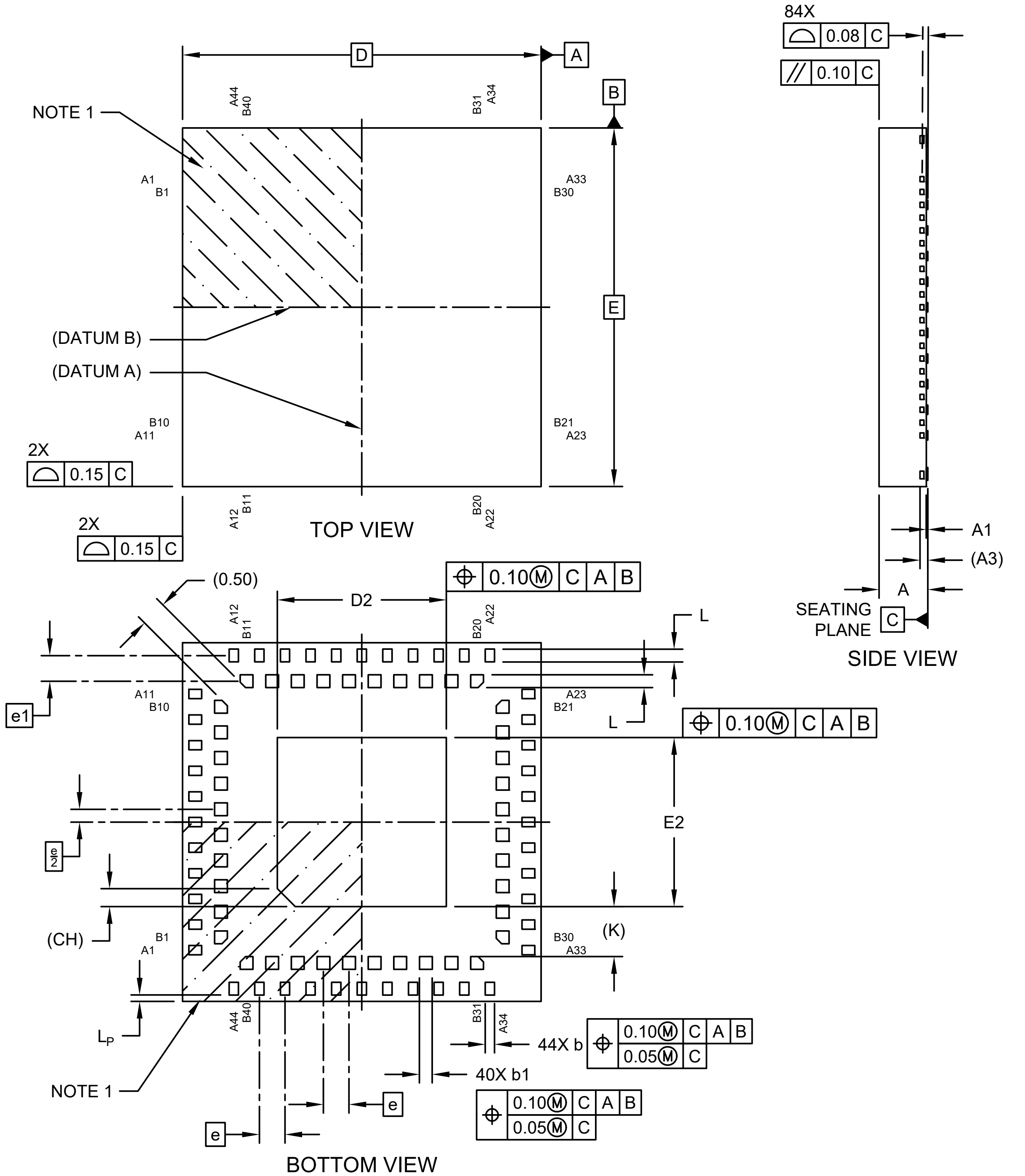
84-Lead Very Thin Quad Flat, No Lead Package (2RW) - 7x7x0.95 mm Body [VQFN] Dual Row with 3.3 mm Exposed Pad

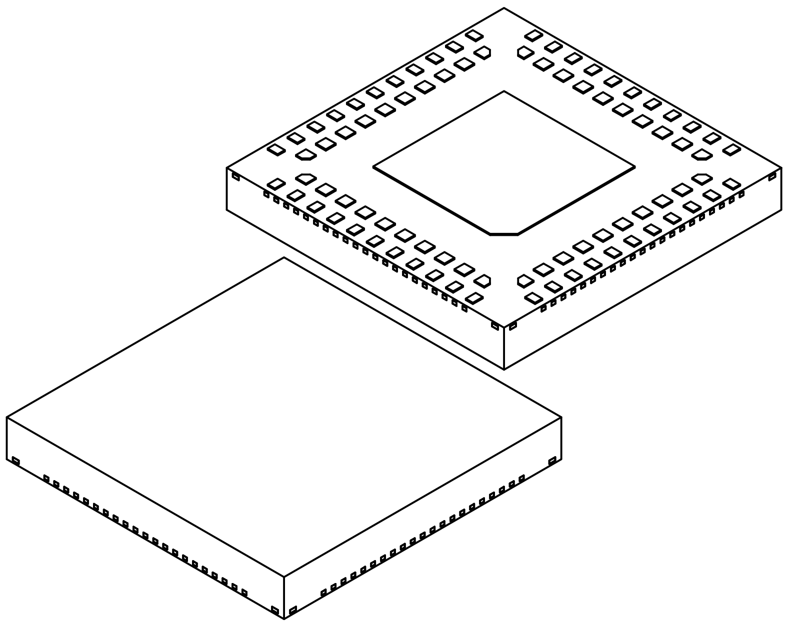
Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-551 Rev A Sheet 1 of 2

84-Lead Very Thin Quad Flat, No Lead Package (2RW) - 7x7x0.95 mm Body [VQFN] Dual Row with 3.3 mm Exposed Pad

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		84	
Pitch	e		0.50 BSC	
Pitch Between Rows	e1		0.50 BSC	
Overall Height	A	0.85	0.90	0.95
Standoff	A1	0.00	0.01	0.03
Terminal Thickness	A3		0.152 REF	
Overall Length	D		7.00 BSC	
Exposed Pad Length	D2	3.20	3.30	3.40
Overall Width	E		7.00 BSC	
Exposed Pad Width	E2	3.20	3.30	3.40
Exposed Pad Index Corner Chamfer	CH		0.35 REF	
Outer Terminal Width	b	0.13	0.18	0.23
Inner Terminal Width	b1	0.20	0.25	0.30
Terminal Length	L	0.20	0.25	0.30
Outer Terminal Pullback	L_P		.125 REF	
Terminal-to-Exposed-Pad	K		0.98 REF	

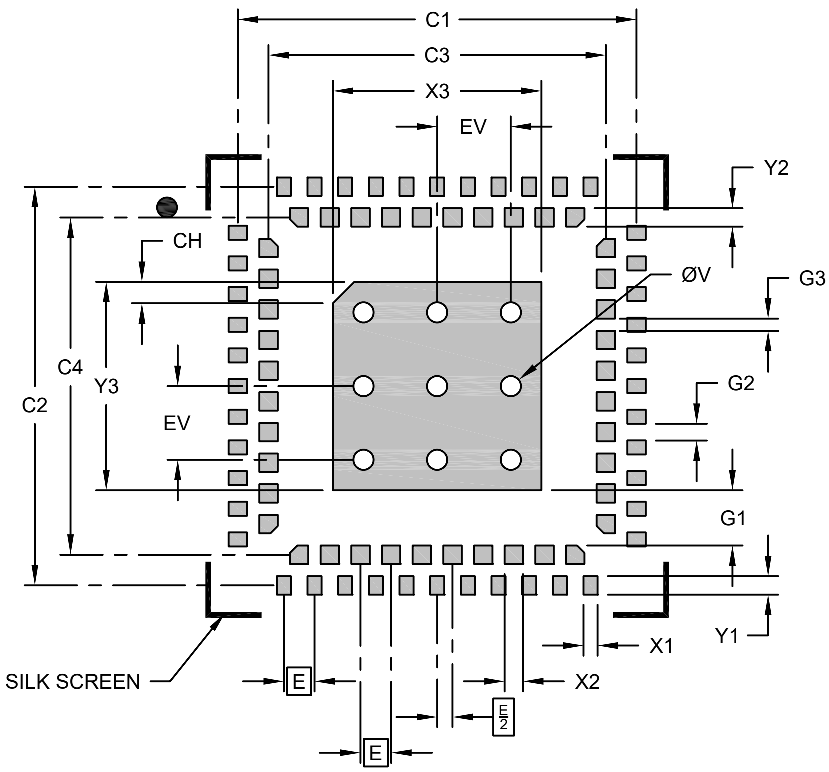
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-551 Rev A Sheet 2 of 2

84-Lead Very Thin Quad Flat, No Lead Package (2RW) - 7x7x0.95 mm Body [VQFN] Dual Row with 3.3 mm Exposed Pad

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Center Pad Width	X3			3.40
Center Pad Length	Y3			3.40
Outer Contact Pad Spacing	C1		6.50	
Outer Contact Pad Spacing	C2		6.50	
Inner Contact Pad Spacing	C3		5.50	
Inner Contact Pad Spacing	C4		5.50	
Outer Contact Pad Width (X44)	X1			0.23
Outer Contact Pad Length (X44)	Y1			0.30

	Units	MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Inner Contact Pad Width (X40)	X2			0.30
Inner Contact Pad Length (X40)	Y2			0.30
Center Pad Corner Chamfer	CH		0.35	
Contact Pad to Center Pad (X40)	G1			0.90
Contact Pad to Contact Pad (X40)	G2			0.27
Contact Pad to Contact Pad (X36)	G3			0.20
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-02551 Rev A